



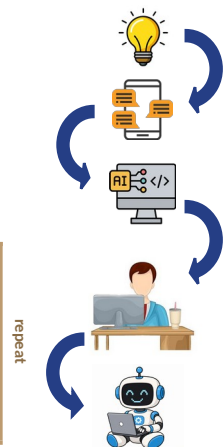
BACKGROUND

In the traditional software development pipeline, software engineering expertise often represents a bottleneck between ideation and a working prototype. Large language models (LLMs) have lowered this barrier, enabling end users, including clinicians, to translate novel ideas into functional applications tailored to their workflows through an emerging practice sometimes termed “**vibe coding**.” The following workflow was used to develop an iOS application providing point-of-care anesthetic reference values.

DEVELOPMENT WORKFLOW

An iterative build loop with the AI assistant

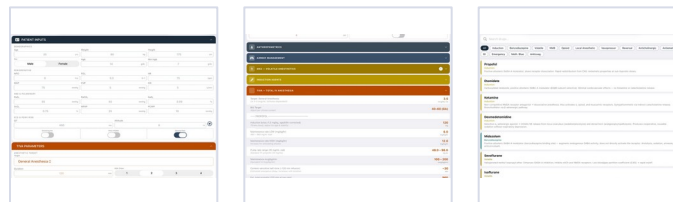
- 1 Think of an application**
User forms the idea
- 2 Define constraints**
User gives the AI requirements
- 3 Create a prototype**
AI builds a first version
- 4 Test the prototype**
Try it, gather feedback
- 5 Define new features**
Identify what to add
- 6 Build the next version**
AI updates the prototype



Steps 4–6 repeat until the prototype meets requirements. No programming training needed — **zero lines of code authored by hand**.

■ User ■ AI

WORKED EXAMPLE — ANESCALC, A NATIVE iOS ANESTHESIA CALCULATOR



Patient inputs

Results dashboard

Drug reference cards

SCOPE — v1.0 SESSION → v2.0 TODAY

116 → **231**

Live calculations

10 → **55**

Drug reference cards

0 → **10**

Clinical tools

0 → **0**

Lines of code authored

FEATURES

Application features

- 16 Clinical domains:** MAC, NMB, TIVA, vasopressors, LAST, OB, peds
- 55 Drug reference cards:** 14 classes, searchable
- 10 Bedside clinical tools:** RSBI, RASS, Fick, driving pressure

100% OFFLINE — no data transmitted

Also: OB/regional protocols · unit conversions · PDF export

Dosing verification audit

191

rows audited

24

sections

3

independent passes

An external reviewer independently audited every calculation. Each entry passed through triage, targeted deep dive, then full independent literature search — cross-referenced against UpToDate, Miller's 9e, and Barash 8e. **All confirmed discrepancies were corrected before version finalized.**

BEYOND THIS APPLICATION

The workflow, not the app, is the transferable product. The same six steps apply to almost any clinical need:

- Custom calculators and order-set helpers inside the EMR
- Desktop widgets for rapid reference at the workstation
- Anesthetic plan generators from structured case data
- Custom case simulators for resident training
- Departmental protocol, checklist, and QI dashboards

LIMITATIONS & FUTURE WORK

Of AI-assisted development, generally

- LLMs reason in language, not graphics — visual and UI design still require human iteration
- Output is bounded by the model's knowledge and the target platform's Software Development Kit
- Hardware and OS compatibility must be verified on real devices
- Institutional IT and security policy may restrict custom software and PHI handling

Of this implementation

- iOS only; not a regulated medical device — every output requires clinician verification
- Final product capability is constrained by the LLM's tooling — user-defined feature scope is bounded by what the model can generate

Future work

- Prospective outcomes validation · FDA SaMD regulatory classification
- Clinician evaluation · formal usability review in training contexts
- App Store sponsorship · Android functionality

AI DISCLOSURE

A conversational LLM, Claude Sonnet 4.6 (Anthropic, PBC), was utilized for native code generation (“vibe coding”). The first author (AP) directed all prompt engineering, designed the system architecture, and verified compilation outputs. **The AI assistant does not hold authorship status.**

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